TRANSFORMATIONAL INTELLIGENCE ∞ RESEARCH EDITION

AGI ALPHA FLYWHEEL

The Enterprise That Invents the Means of Discovering Its Next Economy

Decision-Complete Experimental Instruments:
Construction, Exact Checks, and the Boundary of a Supplied World

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Construct the means. Resolve the decision. Earn the next capability.

Abstract

An autonomous enterprise cannot discover its next economy merely by evaluating more candidates inside an unchanged model. It needs instruments that expose what its present observations cannot decide, and a disciplined account of what an additional construction changes. This chapter builds a small executable instrument for that purpose: given finite candidate worlds, required actions, admissible experiments, costs, and bounded observation errors, it returns either a minimum-cost decision-complete experiment suite or an explicit indistinguishability witness. Its mathematics is a bounded-error specialization of established generalized test-cover design; algorithmic novelty is not claimed. A separately authored evaluator, frozen before the producer saw its tasks, checked 120 fresh synthetic problems. All 82 feasible suites and 38 impossibility witnesses were correct and matched an exact conventional solver. On 79 problems where both action and full-world identification were possible, action identification cost 405 acquisition units versus 659 for full identification. This is work avoided by changing the objective, not superiority over equally formulated conventional practice. Ignoring noise produced 57 false sufficiency claims; omitting one world invalidated 20 of 92 reduced-model suites. Additional experiments exercise a protected, simulated artifact exchange and inspect four repository boundaries. The paper preserves the distinction between a working instrument, a new scientific discovery, a viable market, available cash, and a profit-funded successor. It delivers the first result, useful counterexamples, and reproducible evidence; it does not establish the entire commercial flywheel.

Keywords: experimental design; action identifiability; generalized test cover; bounded uncertainty; autonomous enterprise; evidence; simulated exchange.

1 The research question

The ambition of AGI ALPHA FLYWHEEL is an enterprise whose useful achievements expand the set of valuable activities it can investigate and perform. Its causal chain is specific: discover a limit; construct missing means; enable useful work; deliver accepted value; collect earned resources; build another capability; and test fresh successor work. Each arrow needs evidence. A technically correct program supplies only some of those arrows.

This chapter asks a narrow question within that larger program:

> What must an enterprise observe before it can safely distinguish the actions that matter, and how can a constructed instrument show both a sufficient experiment plan and the cases in which no permitted plan can suffice?

The distinction between identifying a world and identifying an action is economically relevant. Two possible mechanisms may prescribe the same repair, production recipe, or containment action. Fully resolving their difference can consume resources without changing what the beneficiary should do. Conversely, a small unresolved distinction can be decisive when it changes the required action. A useful discovery instrument should expose this difference rather than collapse both cases into a confident recommendation.

The present construction is a *decision-complete experiment compiler*. It accepts an explicit model of possible worlds and returns a plan with an inspectable justification. When the supplied experiment catalogue cannot separate incompatible actions, the output is a pair of worlds and a shared possible observation. That is a constructive research result: it tells the enterprise precisely which claimed decision lacks an observational basis. It does not, by itself, tell the enterprise which new physical observable to invent.

The contribution is an executable, independently checked implementation and an integrated evidence argument. The optimization principle is established. The paper makes its predecessor relationship, strong conventional comparison, failures under model omission, payment controls, source boundaries, and financial limits visible alongside its positive results.

1.1 What was actually constructed

The new source consists of an exact integer-response compiler, a separately authored MILP and exhaustive-search verifier, a frozen fresh-task generator, adverse model-scope tests, a local durable exchange with a controlled byte-release interface, a reproducible example, and a complete requirements register. The producer source was frozen before the fresh tasks were disclosed. The same producer generated the scientific results and the artifact used in the exchange journey. There is no manuscript-only replacement solver behind the reported results.

The publication is a new research contribution staged for an additive Alpha research directory on a pinned base. It is not a claim that the four current repositories have been installed together as a qualified product. Exact existing source modules were inspected and exercised on their supported historical fixture, and the new artifact's unsupported schema boundary was tested directly. Those distinct observations are reported separately in Section 9.

1.2 A claim ladder

The evidence supports finite-model instrument operation and parity with a strong exact conventional formulation. It supports a purpose-relative reduction in acquisition work compared with complete state identification. It also supplies counterexamples to stronger guarantees. It does not establish field performance, a new algorithm, superiority over the predecessor's adaptive policy, a calibrated frontier-model advantage, real demand, complete-cost commercial profitability, or a newly profit-funded successor. These are separate research questions, rather than conditions that can be inferred from the compiler's correctness.

Preceding work	Supported construction	Limit retained here
Transformational Intelligence	Separates purpose, formulation, evaluation, and intervention; a supplied causal task benefits from active observation.	Its unrestricted conventional planner ties; supplied worlds do not establish general discovery.
Invent the Economy	Links technical feasibility, voluntary arrangements, and exact simulated ledgers.	Counterparties, costs, and the technical core are supplied; expected surplus and liquidity differ.
Out-Evolve	Shows how earning a structured response class can change the required identification queries.	Physical structure must be established; known-grammar conventional methods can tie.
Transformational Autonomous Enterprises	Reuses content-bound generic checks while retaining fresh mission checks and separate authority across four source interfaces.	Local-source composition is not production installation; complete cost and market demand remain unresolved.
Adaptive discovery instrument, PR 5012	Constructs minimax observation-conditioned policies and matches its conventional oracle on 24 authored cases.	It uses supplied finite semantics; the stored four-case static gap was deliberately constructed; no qualified downstream market follows.

Table 1: Research continuity and preserved limits. Manuscripts and exact draft/source identities are cited in the text and evidence manifest.

2 Continuity, fresh scope, and the strongest alternatives

The campaign began by reading actual predecessor manuscripts and current source-bound records. Their useful findings were retained without inheriting their application choices. Table 1 records the relevant continuity; historical results in this table are documentary findings from those works, not reruns performed in this campaign.

The required series manuscripts are *Transformational Intelligence*, *Invent the Economy*, *Out-Evolve*, and *Transformational Autonomous Enterprises* [3–6]. A further discoverability contribution note makes the remaining frontier especially explicit: qualifying the semantics of new primitives, noisy partial-purpose synthesis, and actual apparatus cannot be replaced by operating inside a supplied grammar [2]. That is the source-grounded bottleneck motivating this chapter. The present instrument addresses a narrower part of it: identifying which action distinctions the declared means of observation do and do not support. It does not solve the full semantic-construction problem.

The current developer record, Alpha PR 5012 at commit `7931db1f7cf7ceae0d8ae2c3fd3a7c13e9ec16c8`, was inspected rather than inferred from its title. Its instrument builds adaptive policies over finite hypotheses, possible outcomes, costs, depth, and budget. Its stored results report 24 decisions and optimum costs matching a conventional adaptive solver, 12 feasible cases for both, eight for the static comparator, and six malformed-input refusals. Its own uncertainty and downstream HOLD decisions remain valid [9]. A new fixed-suite test must not misrepresent that predecessor as a weak static planner.

2.1 Prior art rules out an algorithmic novelty claim

Generalized test collection already supports heterogeneous test costs and multiple readings associated with the same target class. Its integer-programming formulation separates cross-class reading pairs [8]. Fault-location sensor placement also has a direct minimum-test-cover formulation [20]. The correct strong comparator for this paper is therefore an exact solver with the *same action classes, costs,*

response matrix, and noise model. Full-state identification is a useful secondary comparison because it asks for more information, but it is not the strongest alternative for the action problem.

Goal-oriented optimal experimental design similarly targets quantities needed for downstream prediction rather than complete parameter recovery [1]. Existing physical work on enzymatic reaction control further demonstrates that informative perturbations and iterative experiment selection are already practical research methods [22]. The present compiler's usefulness must come from its executed scope, explicit evidence, and integration fitness, rather than from overlooking those alternatives.

2.2 Why this campaign is timely, and what "why now" does not establish

Current scientific computing and autonomous artifact construction make a tightly checked implementation inexpensive to assemble. Recent OpenAI releases emphasize inspectable outputs, differentiated verification states, corrections, and scrutiny by the relevant research communities [15, 17]. Those practices motivate the evidence discipline here. They do not transfer those sources' scientific claims, model access, or verification status to this work. The supplied computer-use lead was also verified as a current public source, but its product claims were not used as measured performance of this instrument [18].

No temporal discontinuity has been established that makes generalized test cover newly possible. That distinction matters to downstream admission: technical accessibility today is not evidence that a previously unavailable scientific or productive activity has just become feasible.

3 Three consequential downstream possibilities

Before committing to an application, the campaign compared three different productive mechanisms. They are research targets, not three domains on which the abstract test generator has already been validated.

Chemical mechanism discrimination. A production laboratory may need to choose a reaction protocol when different fitted mechanisms recommend different interventions. The missing evidence could be produced by actively changing stimuli, timing, or reaction conditions. A pertinent simulated TAP study shows why nominal discrimination is insufficient: parameter reoptimization can remove the apparent distinction between mechanisms [23]. Existing experimentally demonstrated OED with open code is a strong alternative [22]. A decisive extension would generate feasible perturbations from a physical model, retain parameter-refitting uncertainty, and test fresh mechanisms and observations. This campaign supplies no physical reaction results.

Fault-isolating maintenance. An infrastructure operator may need a targeted repair or isolation action while avoiding unnecessary instrumentation. Here the productive construction changes observation locations or diagnostic tests, rather than intervening in reaction kinetics. Test-cover sensor placement already addresses this problem directly [8, 20]. A useful new offering would have to demonstrate an actual reliability or affordability threshold under sensor loss, changed operating regimes, and unseen fault combinations. The abstract certificates alone do not establish that threshold.

Inverse environmental attribution. A response operator may need to select containment despite confounded source and transport explanations. Strategically timed and located samples could change that decision. Goal-oriented and robust inverse-problem design, as well as integrated source-inversion and response tools, supply substantive existing alternatives [1, 7, 21]. A decisive test needs a validated forward transport model, held-out source/transport combinations, and evidence that the resulting containment action works.

All three remain downstream BLUE SKY HOLD. Their importance is supported; the claimed residual limitation after competent existing practice, physical realization, buyer preference, and newly crossed economic threshold are not yet established by this campaign. The dedicated instrument research allocation therefore continued, while substantial downstream product construction remained closed. A small simulation payment cannot change that admission decision.

4 The decision-complete instrument

4.1 Declared model and scope

Let the finite hypothesis set be $\mathcal{W} = \{1, \dots, n\}$. Each world i has a required action $a_i \in \mathcal{A}$. The admissible experiment catalogue is $\mathcal{E} = \{1, \dots, m\}$. Experiment e costs $c_e > 0$ and has nominal response H_{ie} in world i. Its declared error radius satisfies $\epsilon_e \geq 0$. An executed observation obeys

$$Y_e = H_{ie} + \eta_e, \qquad |\eta_e| \leq \epsilon_e. \tag{1}$$

The uncertainty set is rectangular: every combination of coordinate errors inside the stated intervals is allowed. No probability distribution or independence-in-probability assumption is used. The campaign stores responses, costs, and radii as integers so that separation decisions are exact. The program does not infer H, a, c, or ϵ from physical data.

For a selected suite $S \subseteq \mathcal{E}$, define the possible-observation box

$$B_i(S) = \prod_{e \in S} [H_{ie} - \epsilon_e, H_{ie} + \epsilon_e]. \tag{2}$$

Given an observation y, its compatibility set is

$$K_S(y) = \{i \in \mathcal{W} : y \in B_i(S)\}. \tag{3}$$

A decoder identifies an action only when $K_S(y)$ is nonempty and all its members have the same action. It abstains when multiple actions remain; it flags an observation outside the supplied model when the compatibility set is empty.

Definition 1 (Decision completeness)**.** *A suite S is decision-complete for the declared model if a decoder can return the correct required action for every $i \in \mathcal{W}$ and every error vector permitted by Equation* (1)*.*

4.2 A sufficient suite or an explicit impossibility witness

Proposition 1 (Cross-action separation)**.** *Under the finite-world, known-response, rectangular bounded-error assumptions, S is decision-complete if and only if*

$$\forall i < j \text{ with } a_i \neq a_j, \quad \exists e \in S : \ |H_{ie} - H_{je}| > 2\epsilon_e. \tag{4}$$

Proof. Two one-dimensional closed intervals of equal radius ϵ_e are disjoint exactly when their centers differ by more than $2\epsilon_e$. Cartesian boxes intersect exactly when every coordinate interval intersects. If Equation (4) fails for an incompatible-action pair, their boxes share an observation, which cannot be assigned both required actions by any always-correct decoder. Conversely, if every cross-action pair has disjoint boxes, any nonempty compatibility set contains only one action. Returning that action is correct for every permitted observation. □

The strict inequality is essential. At equality the intervals touch, leaving a valid ambiguous observation. When an incompatible-action pair is not separated even by the entire catalogue, the compiler returns the pair and twice its midpoint vector. The midpoint is permitted in both worlds, and its doubled representation keeps the witness exact even when the midpoint is a half-integer.

This proposition is an elementary bounded-error specialization of the established test-cover structure, included to specify precisely what the software certifies. It is not presented as a newly discovered general theory.

4.3 Minimum-cost construction

Let $P = \{(i, j) : i < j,\ a_i \neq a_j\}$ and set

$$b_{ije} = \mathbf{1}\{|H_{ie} - H_{je}| > 2\epsilon_e\}. \tag{5}$$

The required design is

$$\begin{aligned} \min_{z \in \{0,1\}^m} \quad & \sum_{e=1}^{m} c_e z_e, \\ \text{subject to} \quad & \sum_{e=1}^{m} b_{ije} z_e \geq 1 \quad \forall (i, j) \in P. \end{aligned} \tag{6}$$

The producer uses exhaustive subset enumeration with bit-set coverage, capped at a declared catalogue size. The independent conventional implementation uses SciPy's HiGHS MILP interface and a separately written exhaustive solver [19]. Exact pair checks validate the returned integer suite; a floating-point MILP status alone is not the certificate.

A successful artifact includes the selected indices, cost, input digest, target and noise model, and one separating experiment with its positive margin for every cross-action pair. The digest binds canonical JSON of the four core fields H, actions, costs, and radii. The complete delivery hash additionally binds the artifact schema and result metadata. A constant-action model correctly needs an empty suite and zero incremental acquisition cost. A catalogue containing an unseparable incompatible-action pair receives an impossibility witness, not an arbitrary expensive plan.

The solver has exponential worst-case enumeration cost. The study's $m \leq 14$ is a bounded verification setting, not evidence of production scalability. Conventional MILP, preprocessing, adaptive policies, and richer geometric methods may be preferable at other sizes. The compiler supplies an inspectable reference implementation; it is not a claim to have displaced those methods.

4.4 What processing cannot create

Proposition 2 (An alias cannot be repaired by postprocessing)**.** *If two worlds requiring different actions have the same entire available observation, then any deterministic transformation of that observation also agrees between the worlds. Randomized postprocessing cannot guarantee the correct action in both worlds.*

Proof. Applying the same function to equal inputs produces equal outputs. For a common admissible observation y, a postprocessor whose only input is y, and whose internal randomness is independent of the true world, has the same conditional output law $Q(\cdot \mid y)$. It cannot assign probability one to two distinct required actions. □



Figure 1: Illustrative observation geometry, not a physical measurement. Closed noise intervals that overlap cannot support a guaranteed distinction between different actions. A sufficient additional coordinate can separate the boxes. The certificate remains conditional on the worlds and error bounds included in the model.

The implication is practical. A longer prompt, an extra dashboard, or a transformed representation can improve computation, but cannot remove an observational alias by assertion. The enterprise must acquire a genuinely discriminating observation, establish an additional justified assumption, or preserve abstention. Constructing the means of discovery requires identifying which of those changes is real.

4.5 The boundary of the supplied world

An in-model certificate does not certify model completeness. Add a new world i' with the same response row as an included world i, but $a_{i'} \neq a_i$. No permitted experiment can distinguish them. The old artifact may remain a valid statement about its original scope while becoming insufficient for the expanded task. Requalification must therefore bind the hypothesis set, action semantics, response semantics, and noise envelope, not only the executable's version.

5 Protocol, independence, and resource discipline

The instrument charter preceded construction. It declared the hypothesis, exact comparator, bounded scope, adversarial cases, one fresh confirmation campaign, and protected finalization resources. The aggregate wall-time allowance was 150 minutes: 25 for source diagnosis, 35 for construction, 20 for independent evaluation, 15 for downstream comparison, 10 for the side exchange probe, 15 for integration inspection, and 30 protected for publication. Numerical computation was capped at 600 CPU seconds, the fresh set at 128 tasks, catalogue size at 14 in that set, and side-probe orders at 4,096. Real spending and new paid API calls were zero. No failed result resets an allowance.

Four authored development cases checked a feasible plan, a cross-action alias, a constant-action boundary, and a strict noise boundary. The first run failed an authored expectation: it expected cost two while a valid single cost-one experiment existed. The expectation was corrected after inspection; the producer source did not change. The original failure and explanation are preserved. That correction was completed before any fresh task was seen.

The producer was frozen at SHA-256

```
bda7f986d4e0831a9a21eb22d42212758aba06a6b03aa62989301da3eb3f792f
```

The evaluator was independently written and frozen before disclosure of the producer's final source. It

Family	Tasks	Goal feasible	State feasible	Correct goal	Strict reduction
Nuisance blocks	24	24	23	24	10
Generic random	24	21	21	24	11
Cross-action alias	24	0	0	24	0
Noisy margins	24	13	12	24	7
Simple goal	24	24	23	24	21
Total	120	82	79	120	49

Table 2: One frozen fresh confirmation set. "Correct goal" includes correct impossibility witnesses. Strict reduction counts only problems where both objectives are feasible. Family weights describe test coverage, not field prevalence.

selected 120 tasks, 24 from each of five families, and committed the canonical task set before scoring. The producer and evaluator source hashes were checked after execution. The task commitment and exact reproducibility commands are retained in Appendix A.

5.1 Five fresh task families

The families cover irrelevant nuisance distinctions, generic random response matrices, intrinsic cross-action aliases, strict noisy margins, and simple or constant-action goals. The generic random matrices had full rectangular rank under exact rational elimination; this was checked without filtering or regenerating cases. Equal family weights are an authored coverage design. They are not estimates of the prevalence of these situations in a scientific workload.

The data are synthetic finite diagnostic signatures. They are not measurements of chemical reactions, water networks, or contamination events. The physically distinct possibilities in Section 3 motivated requirements and admission analysis; the five task families tested the abstract contract.

5.2 Comparisons and adverse controls

The primary comparison gives the exact conventional solver the identical model, errors, costs, and action target. A second exact arm asks for complete world identification. A cost-weighted greedy arm tests a practical heuristic. A noise-blind ablation constructs plans as if all error radii were zero and then checks them against the actual radii. Model-scope stresses remove a preselected world or add an observationally identical world requiring another action. Seven artifact mutations test that the checker rejects changed cost, incomplete or invalid suites, false impossibility, and corrupted certificate content.

The confirmation is fresh relative to producer development, but independence has a specific scope: different code and delegated roles under one controller and host. It is not external replication, a hostile isolation boundary, or institutional peer review. There was no direct frontier-model or prompt-only head-to-head benchmark, no new learned specialist, and no model training. Those comparisons remain unrun rather than being assigned perfect surrogate performance.

6 Fresh confirmation results

All 120 action results passed the independent semantic, cost, and optimality checks: 82 sufficient suites and 38 impossibility witnesses. Both independent conventional oracles agreed on feasibility and

minimum cost. The full-world arm also returned 120 correct decisions, with 79 feasible cases. There were no execution errors during confirmation and no post-result producer changes.

Among the 79 jointly feasible tasks, action identification required 405 acquisition units versus 659 for complete state identification, a reduction of $254/659 \approx 38.54\%$ in aggregate. Forty-nine cases were strictly cheaper and 30 tied. Three further tasks allowed the action to be decided even though the full world could not be identified. No infinite saving is assigned to those three cases. Their result is feasibility under a different objective.

The exact action-conditioned library baseline matched the producer on every problem. Thus the 38.54% reduction is *not* a speedup or an algorithmic superiority claim. It measures distinctions no longer required when the target is the action. The distinction between the objective and the implementation is central to interpreting this result.

On all 82 feasible action tasks, the greedy suites were sound. Sixty-three were optimal and 19 cost more than the optimum; aggregate greedy cost was 471 units versus 419. Its 38 infeasible status decisions agreed with the oracle but did not carry impossibility witnesses. They are correctly classified statuses with unsupported proof output, not 38 wrong classifications.

The noise-blind arm falsely reported sufficiency on 57 of 120 tasks when checked against the true declared radii. The distribution was 16 nuisance cases, 19 generic cases, and 22 noisy-margin cases. The alias and simple-goal families added none. This demonstrates a meaningful acceptance failure from an unjustified observation model even when the solver is exact for its own relaxed model.

Removing one preselected world yielded 92 reduced models with a sufficient suite; 20 suites failed when assessed against the original full model. In the separate, deliberately adversarial alias construction, each of the 82 feasible original tasks was expanded with an observational duplicate requiring a different action. All 82 old certificates became insufficient for the expanded problem. Recompilation correctly refused all 82 and supplied an impossibility witness. This does not refute their original scoped guarantees; it refutes silent transfer to a broader model.

All seven artifact mutations were rejected. All 360 input bindings across the goal, full-world, and noise-blind producer arms matched the prescribed canonical input. These are contract and provenance checks, not evidence that supplied physical semantics are true.

6.1 Runtime and resources

The confirmation interval measured 1.004 seconds of wall time and 2.805 seconds of process CPU with native solver parallelism. The 120 action calls totaled 38.68 milliseconds, full-world calls 35.61 milliseconds, greedy calls 6.19 milliseconds, and noise-blind calls 34.16 milliseconds. The paired oracle interval totaled 775.72 milliseconds, including imports and two objectives. These are one ordered local pass and are retained for reproducibility. They are not a controlled latency contest, a model-token bill, a complete delivery cost, or an AI-versus-human labor comparison.

The numerical environment was Python 3.12.14, NumPy 2.3.5, and SciPy 1.17.0 on Linux x86-64. The compiler uses the standard library; the independent conventional oracle uses the installed scientific runtime. No new paid model call was used to produce the numerical evidence.

7 From a checked result to a protected delivery

A useful artifact and an earned collection are different events. The side experiment therefore implements a transaction boundary around the actual scientific output rather than treating an accepted flag as

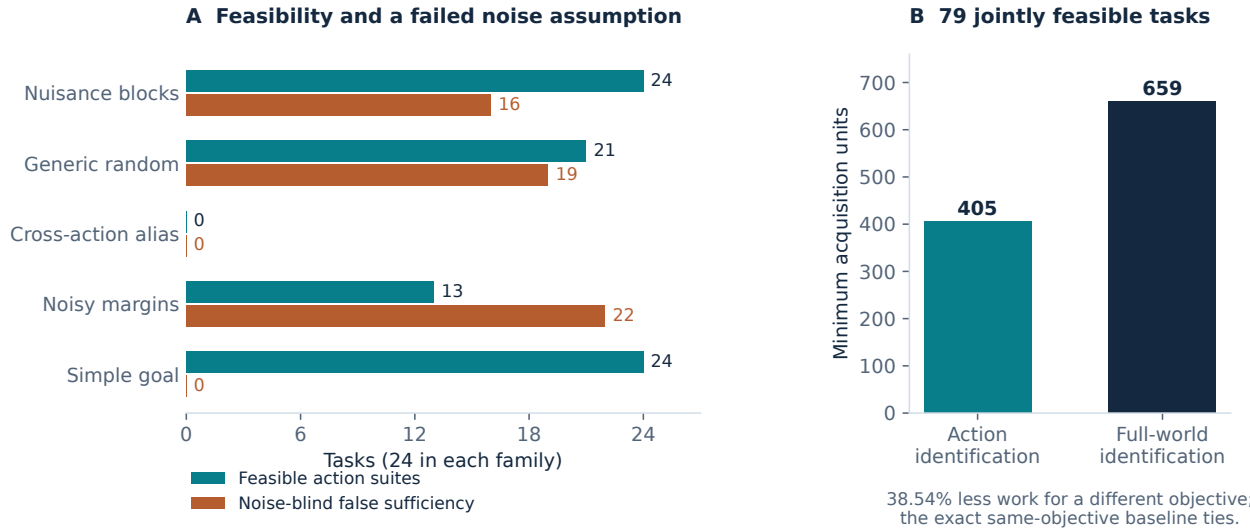


Figure 2: Executed finite-model results. Left: feasible action problems and false sufficiency from ignoring noise, by family. Right: aggregate minimum acquisition cost on the 79 jointly feasible problems. The right panel compares different objectives; the exact action-conditioned library solver ties the compiler.

payment. It is a trusted local Python/SQLite simulation, with integer simulated USD cents. It does not represent live USDC, AGIALPHA, a bank account, or an authenticated remote service.

The order binds immutable scope, price, fee, and the expected artifact digest. The worker supplies the actual bytes. A separately configured checker invokes a deterministic verification callback against the bytes and agreed scope. The callback for the scientific journey binds the task to the order and uses the independent evaluator to check the delivered design. A seller assertion is not sufficient, and a strategic buyer refusal does not override valid checker evidence.

Before settlement, the public `get_artifact()` interface withholds the complete output. Funding moves available simulated buyer money into restricted escrow. Successful private production and verification enable settlement; one SQLite transaction records the earned transfer and buyer entitlement. Interruption before commit rolls back the economic effects. A later reconnect can retrieve the committed entitlement. This is local database atomicity under the tested process model, not distributed atomicity or irreversible payment finality.

The engine also supports an explicit credit exception with eligibility and aggregate exposure limits. Receivables remain distinct from collected cash. Credit defaults, late settlement, chargebacks, refunds, and unpaid refund liabilities preserve costs and obligations. A refunded artifact may already have been learned by the buyer; reversing the financial entry does not claim to reverse knowledge possession.

7.1 Executed adverse paths and a preserved correction

The frozen corrected exchange passed 22 local behavioral tests. They exercised insufficient buyer funding, forged external receipts, semantic and artifact-hash failures, release withholding, replay idempotence, conflicting event keys, a settlement/refund race through separate SQLite connections, injected precommit interruption and restart, settlement delay, refunds, chargebacks, capped credit, partial collections, default, and separate configured worker/checker roles. The tests bind actual bytes and durable state rather than only asserting a success label.

The first suite passed 20 tests but lacked one important case. Review added a delayed-refund regression,

which exposed a defect: after a refund liability was paid later, net collections still showed 925 simulated cents instead of zero. The correction updates later refund/chargeback principal payments and closes the refund-due state after full repayment. The original source, failed targeted test, original database state, correction record, and final run are retained. A separate full-suite attempt interrupted by global storage exhaustion is explicitly preserved as incomplete, not counted as a passing run.

This correction illustrates why an aggregate pass count cannot substitute for an examined accounting contract. The completed suite checks local behaviors; it does not establish hostile-process security, authenticated external identity, untested API/cache/log release boundaries, empirical participant trust, or the full enterprise regression suite.

7.2 One actual scientific certificate through the exchange

The replay selected `generic_random-013`, the first sufficient task in the frozen evaluation order. The compiler selected experiment 6 at an acquisition cost of two units. Its 1,035-byte payload passed the independent scientific verifier, exact task binding, and certificate-witness checks. A pre-funding request for the full bytes was refused. A synthetic buyer then funded 500 cents; settlement charged a declared 25-cent fee and created an entitlement to the exact accepted bytes. Duplicate settlement had no second economic effect, and closing and reopening the database preserved retrieval.

A second order expected the hash of a deliberately mutated empty-suite result. The bytes therefore passed that order's hash condition but failed independent semantic acceptance. The order was rejected and refunded. This distinguishes content binding from scientific validity. The event chain and money conservation checks passed. The successful order's 475-cent remainder is a partial ledger surplus before unknown construction and operating costs, not complete-cost profit.

The original accepted artifact hash is `30a209f8d93e579f9f898d9b22fad2ee49fb987d0105dd79f8fd7ba40c3d170f`. Replayed artifacts can have different full hashes because measured elapsed time is included; the canonical scientific input and checked semantic result remain separately bound.

8 Commercial interpretation and financial discipline

8.1 Buyer participation must survive the ordinary-library alternative

The accepted deliverable is a model-conditional experiment plan and its checkable certificate. Its value comes from the beneficiary's changed decision or avoided acquisition work. A free, competent solver can produce an equivalent plan when the user already has a valid model and can complete the workflow. A price difference between an expensive human service and a short program does not compel anyone to buy.

Let V denote the expected value of the same accepted outcome, P the offered service price, B the buyer's completion burden, L its liquidity/lock-up cost, and R its residual trust and failure cost. Relative to an outside option with completion cost B_0 and risk cost R_0, participation requires

$$V - P - B - L - R \;\geq\; V - B_0 - R_0, \quad \text{hence} \quad P + B + L + R \leq B_0 + R_0, \tag{7}$$

as well as any required funding capacity. When outcomes and risks match and the free alternative is effortless, a positive price cannot win on this condition. In other segments, completing the inputs, checking their meaning, or delivering an accepted result may impose real effort. This paper has not measured that effort for actual buyers and does not invent it to rescue demand.

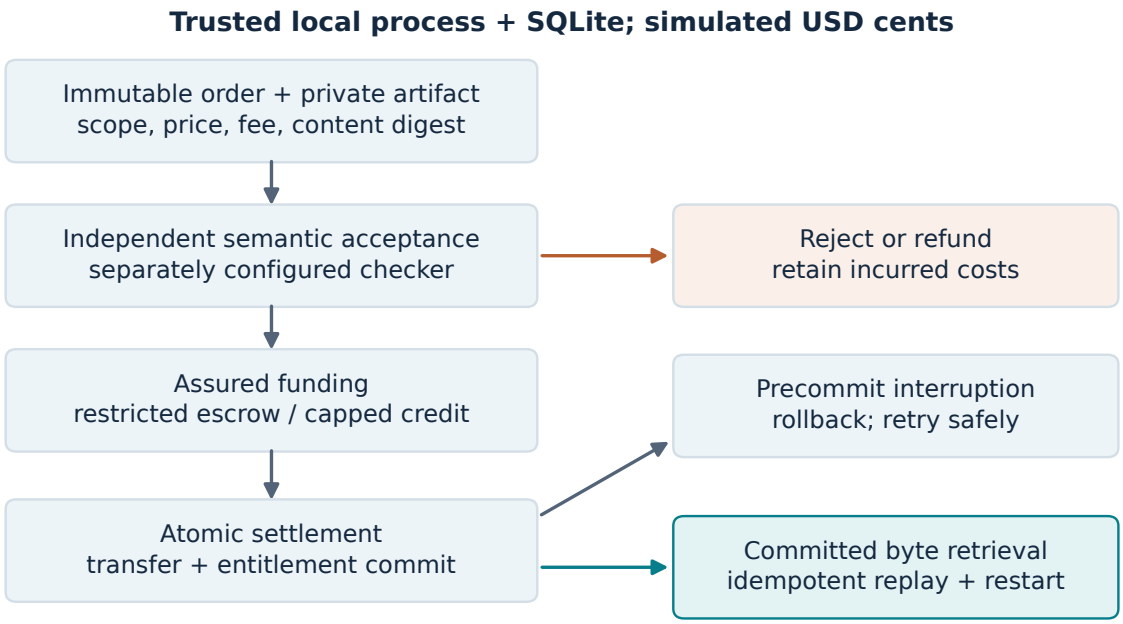


Figure 3: The implemented local exchange boundary. Full useful bytes remain private until committed entitlement, except for the separately capped credit path. Refund, interruption, and reversal controls retain obligations. External payment rails, remote authentication, and four-repository paid execution are not implied by this diagram.

Outside completion effort	Purchases / 64	Gross receipts	Fees	Remainder
$0	0	$0	$0	$0
$3	0	$0	$0	$0
$10	32	$160	$8	$152

Table 3: Hypothetical participation sensitivity, in simulated dollars. Remainder excludes unknown full costs and is not profit. These are analytical consequences of stated assumptions, not 32 executed market sales or calibrated demand.

A frozen analytical sensitivity considered 64 hypothetical buyers with cash budgets of $0, $3, $5, and $10 in equal numbers, a $5 service price, and a declared $0.25 seller settlement fee. The free alternative's residual completion effort was stipulated as $0, $3, or $10. This was a comparison conditional on wanting the same completed task; whether the task is worth undertaking was not estimated. Ties chose the free alternative.

Fund-before-production and verify-preview-then-fund variants tied because no differential trust, delay, or completion burden was assigned. That tie does not establish that actual buyers are indifferent. The effortless equivalent free option attracted no purchases; the favorable $10 effort case only shows the participation condition under that assumed burden and the supplied cash constraints. It cannot qualify Commercial GO.

A research instrument that exposes an ambiguity may be useful even if it makes a proposed sale less attractive. Correct refusal prevents the enterprise from confusing a plausible answer with a deliverable that meets its own contract. That benefit still needs a beneficiary and a justified economic value before it becomes a commercial claim.

8.2 Quantities times rates, and the unresolved full bill

The scientific run records executed CPU and wall time, source identities, task counts, and artifact sizes. The exchange records simulated cash movements, fees, and cost events. The campaign did not receive an itemized bill for all model context, hidden inference, tool use, authoring, review, or host resources. These unknown material quantities are not assigned zero.

As a transparent replacement-cost example, the official standard short-context table consulted on October 28 lists input/output prices per million tokens of US$10/US$50 for GPT-6 Astra, US$2/US$10 for GPT-6.1 Sol, US$1/US$5 for GPT-6.1 Terra, and US$0.10/US$0.50 for GPT-6 Luna [16]. These are quoted tariff scenarios, not evidence that any such billed call occurred. Context length, caching, service mode, and region affect applicability. One may compute $q_{\text{in}} r_{\text{in}} + q_{\text{out}} r_{\text{out}}$ for a declared usage scenario, but the tariff cannot recover missing usage.

The evidence ledger distinguishes measured resource quantities, explicit simulation quantities, and unresolved quantities. Acquisition units in the mathematical tasks are supplied positive integers; they are neither measured laboratory costs nor monetary income. A defensible full-delivery comparison would additionally need model preparation, calibration, semantic validation, retries, support, storage, mechanism operation, and renewal. Neither the current CPU measurement nor a few successful simulated orders accounts for all of those activities.

No sourced equivalent completed human service, validated task-hour requirement, or actual buyer-payment dataset was obtained for this specific deliverable. Consequently the paper does not report an AI-versus-human economic advantage or a calibrated demand estimate. This is a missing comparison with a defined evidence requirement, not a reason to invent salaries or agency overhead for an autonomous process.

8.3 Profit, cash, payback, and funding are separate states

The following quantities must be maintained separately:

$$\Pi = \text{earned revenue} - \text{recognized expenses}, \tag{8}$$

$$C_{\text{free}} = \text{seller cash} - \text{nonoverlapping obligations and restrictions}, \tag{9}$$

$$\text{Payback} = \text{cumulative attributable returns relative to original outlays}, \tag{10}$$

$$\text{Funding provenance} \in \{\text{earned collections, opening capital, debt, mixture}\}. \tag{11}$$

An escrow deposit is a restricted customer balance, not earned seller income. An unpaid receivable can support earned revenue without supplying investable cash. Paying a previously recognized liability changes cash and outstanding obligations without creating the same expense again.

The executed R06 fixture begins with zero seller cash, earns 1,000 ledger units, incurs expenses of 600, collects 650, and pays the 600 expenses. Profit is 400, cash 50, and receivables 350. Nonoverlapping reserves of 20 and working capital of 30 leave no unrestricted cash. A proposed investment of 200 is deferred with a 200 shortfall. Profit remains 400. The code's amounts are integer simulated cents; these illustrative unit choices have no bearing on the identity being tested.

A second executed fixture holds 1,000 in restricted escrow for unsettled work with no earned transfer or buyer entitlement, while seller cash is 100, unpaid obligations are 70, and reserves are 30. Earned revenue from that order is zero and unrestricted cash is zero. The same 200 investment remains unaffordable. The system does not finance construction from customer custody.

A chargeback fixture further shows that accepted work does not make reversal exposure disappear. A 1,000 sale with a 25 fee later experiences a full chargeback and another 15 fee. The specified

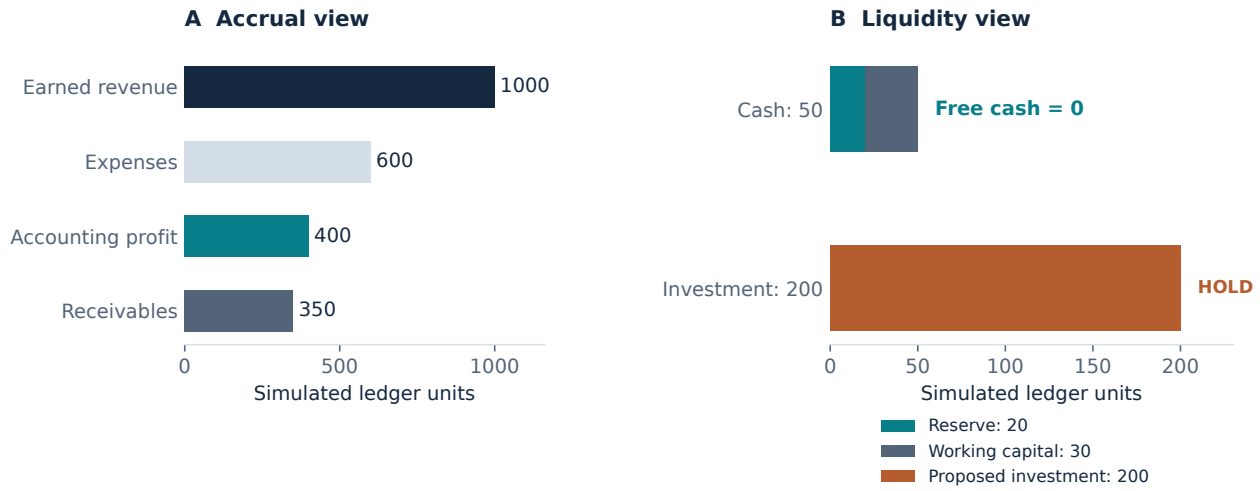


Figure 4: Executed liquidity regression, in simulated ledger units. Positive accounting profit survives an investment deferral: receivables are not collected cash, and the 50 cash units are fully required by the declared nonoverlapping reserve and working-capital amounts.

Repository	Main commit inspected
AGI-Alpha-Agent-v0	`12770c22b4a9f90f2013bbfb64242f912ba5071e`
AGI-Fleet-Linux	`45759fb096faf8988f58ae5ea71248d0f35a97c7`
AGIJobsv0	`bc4abbd5c28f3672275b13300e40ea26c820b8ee`
AGIJobManager	`5d5670df9dbbc79da115e0a44b71989f34df58c1`

Table 4: Exact source identities observed during preflight. Downloaded file digests and URLs are in the evidence package.

accounting treatment preserves earned revenue, records total expenses of 1,040, and produces a loss of 40. Any cash shortage becomes a liability; later payment does not incur the expense twice. These are executed synthetic accounting examples, not observed business performance.

8.4 No profit-funded successor is inferred

The compiler, verifier, and exchange were constructed from the campaign's initial research allocation. Later simulated receipts cannot retrospectively finance work already performed. No new capability was subsequently constructed from reconciled unrestricted earned cash and tested on fresh successor work. Loading the compiler again, reusing its certificate, or increasing a capacity counter would not establish that chain. Full development cost and market participation also remain unresolved, so complete-cost Commercial GO and the completed profitable flywheel are not claimed.

9 Four-repository source inspection and the unsupported edge

The campaign inspected governance, current public source identities, coordination, active ownership, security, and relevant native interfaces in all four repositories. Table 4 records the independently retrieved main revisions. These pins are an inspected source set, not a qualified compatible release.

The active ownership record reserves final integration and release work to the existing primary lane. Concurrent discovery and integration work has existing owners. The fresh contribution is packaged as

an unapplied patch for a uniquely named research folder; no concurrent implementation, release ref, workflow, or publication state was changed [10–14].

Twelve supported source-contract checks passed on Node v24.22.1. They invoked the real Alpha source-selector inspector on its historical fixture, both Jobs and Manager inert receivers with that inspector, refusals of stale context and Boolean-verdict substitutions, refusal of positive drafts on the fixture's `none` route, refusal of an authority-expansion field, and Fleet receipt binding before and after delivery-byte alteration. The nonzero drafts failed at route selection; those outcomes do not independently prove every spend-cap boundary.

The current inspector accepts a closed transformation-envelope schema. A direct attempt to submit the new `agi-alpha.decision-certificate.v1` envelope was rejected with *Unsupported transformation envelope fields*. That is an executed unsupported-interface finding, not a fabricated successful native handoff. The existing current-main receivers cannot be made compatible merely by renaming the new artifact.

The existing native routes retain distinct AGIALPHA and USDC identities and zero allowances in the inspected historical charter. Neither route executed a positive paid job in this campaign. The separate USD-cent exchange is explicitly a local technical simulation and cannot be counted as settlement on either native route. A common reporting unit would not grant conversion, shared balances, or serial double settlement.

Fleet main supplies a receipt byte-binding seam; pending richer receivers remain in unmerged owner branches. Byte binding does not establish a qualified executor, independently authorized checker, or scientific truth. Clean installed-package operation, new-instrument native dispatch, four-repository acceptance and return evidence, route-specific positive settlement, and release qualification remain incomplete. The inspected source snapshots and local tests are preserved as useful partial evidence.

10 Interpretation: what expanded, and what did not

The instrument makes a precise class of investigations executable and inspectable: it determines whether declared observations can support a required action, constructs the least-cost fixed suite when one exists, and exposes a counterexample otherwise. It supplies a reusable artifact with exact input and result binding, independent checks, and a reproducible buyer-facing release boundary. Those are positive engineering results.

The fresh evidence does not establish discovery improvement over the strongest existing action-conditioned design practice. That comparator ties exactly. Nor does a fixed suite establish superiority over the recent adaptive instrument. The measured reduction against full identification arises from asking the relevant question, which is useful but already understood in generalized test collection and goal-oriented design. A stronger scientific claim would require an additional instrument that changes the available observations or reliably qualifies previously unavailable semantics, followed by a matched test of the new investigative action.

The omitted-world stresses sharpen that requirement. Certificates should be transported with their scope and checked again when the world model, error envelope, objective, or permitted actions changes. If a newly considered world is observationally indistinguishable but requires another action, the correct successor behavior is refusal or a genuinely new observation. Retained confidence alone is not an inherited productive advantage.

The commercial interpretation is equally specific. A checked plan can be valuable when it changes a beneficiary's action or avoids consequential acquisition costs. A conventional method can deliver that value profitably. However, conventional parity also makes free completed-service alternatives relevant,

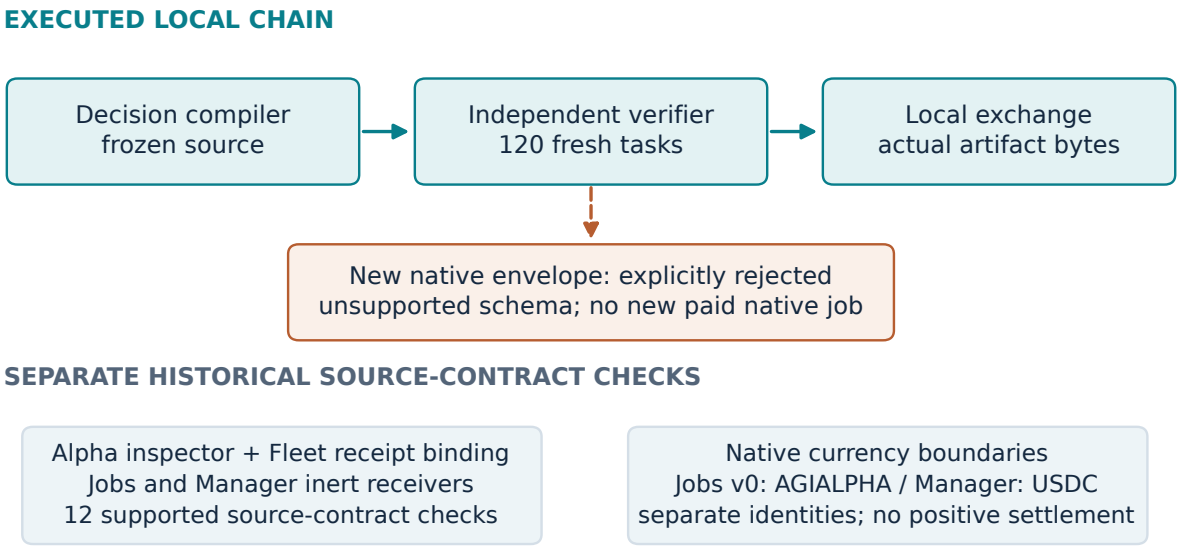


Figure 5: Evidence and authority boundaries. Solid paths represent this campaign's executed local scientific and exchange artifacts. The native source-contract smoke uses its historical fixture. The new instrument's edge into the closed native envelope is explicitly unsupported; dashed paths are required future work, not implemented integration. The two native currency routes remain alternatives with separate identities.

and the present study has no calibrated proof of a paying segment. Payment controls protect declared exchanges; they do not manufacture participation or turn restricted balances into investable profit.

This leaves a concrete next construction question: can an enterprise acquire and validate a new observation or experimental primitive that separates a consequential unresolved action pair under realistic physical and economic constraints? That question follows from the witness, rather than from a preference to report more GO decisions. Its answer must include feasible apparatus or intervention, calibration, model-misspecification checks, beneficiary acceptance, costs, and funded operation.

11 Limitations and completion states

The principal limits are finite supplied worlds, known response semantics, rectangular bounded error, small catalogues, positive fixed acquisition costs, and fixed nonadaptive suites. The worlds and action labels were authored; no domain forward model, laboratory calibration, learned specialist, direct frontier-model benchmark, or real customer study was executed. The exact certificate solves its declared model and does not validate those premises.

Independent code, role separation, mutation tests, exact arithmetic, and two conventional solution methods reduce several implementation risks. They do not provide external replication, isolation from a malicious host, or a theorem-prover formalization. The exchange assumes a trusted local process and database. Its atomicity and authority checks have that same scope.

All 52 inherited and added requirements are retained in the coverage register. Most are broader than this bounded instrument experiment. Appendix C identifies local passes, partial evidence, and unrun

Decision	Evidence-based state
Bounded instrument construction	Completed under the initial research allocation.
Finite-model operation	Supported: 120/120 independently checked, exact-library parity.
Purpose-relative acquisition reduction	Supported against full-world identification, a different objective.
New algorithm / strongest-baseline advantage	Not established.
Open-world discovery improvement	Not established; explicit scope stresses fail transfer.
Downstream Blue Sky flagship	HOLD for all three researched possibilities.
Local exchange controls	Bounded support from 22 corrected tests and actual artifact replay.
Calibrated participation / full-cost profitability	Unresolved; sensitivity is not demand evidence and the full bill is unknown.
Unrestricted profit-funded construction	Not demonstrated.
Installed four-repository operation	Incomplete; source smoke is partial and the new schema is unsupported.
Release / deployment / public publication	Not performed; the deliverable is a research edition and reproducibility package.

Table 5: Independent completion states. A positive result at one stage does not imply another.

requirements individually. The 120 scientific cases, 22 local exchange tests, and 12 source-contract checks must not be added together and advertised as 52 enterprise regressions passed.

12 Conclusion

Constructing the means of discovery starts with identifying what a decision requires and what the present observations cannot supply. This campaign built and checked an instrument that makes that boundary explicit. It returned correct least-cost plans or impossibility witnesses on a fresh, frozen set, matched a strong conventional solver, and exposed how unjustified noise assumptions and missing worlds can break a stronger claim. It also carried a real computed artifact through a protected local exchange while preserving financial and source boundaries.

The next economy has not been discovered merely because those components operate. The new paper's result is a concrete investigative capability, a rigorous statement of its scope, and a reproducible account of the remaining causal links. Progress toward the larger enterprise requires a newly justified observation or productive arrangement, accepted beneficiary value, complete costs, earned unrestricted collections, and fresh use of a subsequently constructed capability. The ambition remains the enterprise that builds its next frontier; the evidence remains what this edition actually constructs and tests.

Research provenance and availability

Edition `AAFW-DCE-2026-10-28-r1` was prepared with AI-assisted research, code construction, and drafting, under the stated Vincent Boucher publication identity. Scientific computations were executed using the included frozen sources. All financial activity is simulated. No new paid API call, purchase, customer contact, real transfer, repository mutation, public release, or deployment was performed. The companion package contains source, bibliography, figures, raw compressed observations, original failures and corrections, exact-source records, and tested reproduction commands. Browser and PDF exports represent the same research edition. Private author-provided manuscripts are identified as

such; citations do not imply peer-reviewed or public release status.

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A Reproduction, source identities, and the user journey

The package separates `src/`, `examples/`, `evidence/`, `sources/`, and `paper/`. The primary compiler requires Python's standard library. The independently authored oracle requires NumPy and SciPy; the tested versions are recorded in Section 6. The proposed contribution carries Apache-2.0 headers; frozen executed source bytes remain preserved unchanged and separately hashed. The additive patch targets a new `research/decision_complete_discovery/` directory on the pinned Alpha base; applying it is not a merge or release.

The tested local workflow is:

1. Read the task's world set, action target, observation responses, costs, and noise radii. These are substantive assumptions.
2. Run the compiler on the example task to obtain the suite or witness.
3. Invoke the independent verifier and inspect exact pair margins and the agreed minimum cost.
4. Run the scientific-artifact exchange journey. Observe that useful bytes are withheld before settlement and match the accepted digest after committed entitlement.
5. Inspect costs, restricted balances, liquidity decisions, and the unsupported native-interface record.
6. Reproduce the frozen confirmation and local behavioral checks. Preserve the distinction between a rerun of this known set and a new fresh evaluation.

Run these commands from the extracted package root with the declared runtime available. Each output path must be new, preserving the supplied original evidence.

```
python src/decision_instrument.py examples/scientific_task.json \
  --output reproduced_certificate.json
python src/independent_evaluation.py selfcheck
python src/replay.py confirmation --output replay_confirmation
python src/replay.py exchange --output replay_exchange
python src/exchange_checks.py --output replay_exchange_checks
python src/replay.py native --output replay_native
```

The portable replay wrapper is separately identified and does not modify the frozen producer or evaluator. The original `independent_run.py` is retained as the historical driver with its original campaign path; it is not advertised as a portable entry point. The confirmation replay reproduces known published tasks and must never be relabeled fresh confirmation. Expected core totals are 120 correct decisions, 82 feasible suites, 38 impossibility witnesses, and jointly feasible acquisition totals 405 and 659. Timing values and hashes containing timing metadata can vary.

For the paper, run `python src/make_figures.py`, change to `paper/`, then run `pdflatex`, `bibtex`, and `pdflatex` twice on `AGI_ALPHA_FLYWHEEL_Decision_Complete`. The supplied figures and bibliography are part of the source package. The README gives exact shell commands and the tested versions.

The producer's input digest binds sorted compact JSON of `H`, `actions`, `costs`, and `eps`. Metadata such as task names does not change the mathematical input binding. Complete payload hashing covers the delivery envelope. The independent evaluator hash is

`ba8210bbc6735f73f7de7ba7838d3564d1c57616dc195037b75b6e1330fb7f96`

and the canonical fresh task-set commitment is

`263362843733f69f71d9eea39433a05059dbc5cb1d473117bbafe6ca453e7041.`

Hashes, commands, byte counts, exit codes, runtime identities, and compressed database traces are available in the machine-readable manifests. The package's downloaded source snapshots are an intentionally bounded subset of the inspected repositories; their source URLs and exact digests are retained. They are not a complete checkout or dependency lock for a production installation.

B Evidence interpretation and resource ledger

Activity or quantity	Measurement class	Interpretation
Producer and oracle execution	Measured local CPU / wall time	Numerical verification cost only; no conversion to human working hours.
Acquisition cost in test design	Supplied integer units	Objective coefficients; not measured physical expense or money.
Artifact production and release	Executed bytes / digests / events	Local work and entitlement evidence; no physical scientific outcome.
Exchange payments and fees	Explicit simulated quantities	Integer simulated USD cents; no live token or bank transfer.
Buyer completion-effort scenarios	Sensitivity assumptions	Same-outcome outside option; not empirically calibrated demand.
Model API benchmark rates	Sourced public tariff	Counterfactual replacement-cost input, not this session's observed bill.
Authoring and total model/tool usage	Unresolved material quantity	Not silently zero; prevents complete-cost commercial qualification.
Seed research allocation	Existing authorized resources	Instrument construction precedes any simulated receipt; no retrospective profit-funding claim.
Native source contracts	Executed historical-fixture smoke	Source-module behavior only; new instrument's native edge remains unsupported.
External installation / release	Not executed	No qualification inherited from a predecessor or another revision.

The root execution filesystem initially had no available disk space. The campaign used a narrowly scoped temporary workspace, compressed its own evidence with byte-for-byte verification, and preserved a durable recovery checkpoint. Other tasks' files were not deleted. Original setup errors, a storage-interrupted test attempt, and the accounting correction remain in the package. Temporary capacity constraints do not change the scientific acceptance rule or convert missing results into passes.

The original charter also capped persistent temporary data at 4,400,000 bytes and temporary rendering at 600,000 bytes. Those small finalization ceilings were not maintained: the finalization record retains measured workspace checkpoints and the subsequent render inventory. Moving to a private build area did not reset the limits. This is a disclosed publication-resource overrun, not evidence of budget compliance. The 600-CPU-second numerical cap is distinct from document construction and inspection; the measured fresh numerical run used 2.805 CPU seconds. Full campaign inference, tool and host consumption, as well as complete per-stage timing, remain unmetered. No complete-cost conclusion follows from the numerical run alone.

C All 52 retained requirements and their coverage

The register below retains R01–R37 and R38–R52. PASS LOCAL means that a specifically named local fixture passed on identified source bytes; the row states its narrow scope. It is not a full pass for the parent requirement. In particular, R28 covers a restricted-cash checkpoint after private production and verification, not the entire specified pre-work and post-release sequence. PARTIAL identifies a relevant executed component or documentary evidence without claiming the complete requirement. NOT RUN means that the required integrated behavior has not been established. The full register retains summaries, evidence pointers, and remaining qualification limits. A paper's discussion of a requirement is not an executed regression.

ID	Coverage	Requirement and exact qualification limit
R01	NOT RUN	**Technically conventional, commercially profitable.** Permit Commercial GO for a profitable conventional offering despite technical parity; prohibit unsupported technical-superiority claims. *Scope:* No corresponding regression result attached; module existence is not a pass.
R02	NOT RUN	**Technically impressive, commercially losing.** Preserve a technical win while withholding commercial profitability when complete enterprise costs exceed revenue. *Scope:* No corresponding regression result attached; module existence is not a pass.
R03	NOT RUN	**Profitable base business, uneconomic specialist.** Retain a profitable base offering and reject an optional specialist whose incremental value fails to cover retention cost. *Scope:* No corresponding regression result attached; module existence is not a pass.
R04	NOT RUN	**Positive cash, negative profit.** Distinguish cash from equity, debt, deposits or timing from profit; positive cash cannot establish profitability. *Scope:* No corresponding regression result attached; module existence is not a pass.
R05	NOT RUN	**Positive profit, incomplete project payback.** Allow justified cash-affordable investment before initial project payback while reporting unrecovered outlays and funding provenance. *Scope:* No corresponding regression result attached; module existence is not a pass.
R06	PASS LOCAL	**Positive accounting profit, insufficient unrestricted cash.** Execute profit400/cash50/receivables350 fixture; reserve50 and defer investment200 for liquidity without erasing profit. *Scope:* Exact local integer fixture: earned revenue 1000, expenses 600, collections 650, cash paid 600, profit 400, cash 50, receivables 350. Reserve 20 plus working capital 30 leave free cash 0. Investment 200 is held with shortfall 200 and no construction. Trusted local simulated cents; no enterprise profitability or native investment qualification.
R07	PARTIAL	**Unfunded liability.** Retain incurred liabilities and appropriate expenses when immediate payment is impossible. *Scope:* Local fixtures preserve unpaid preview expense and refund obligations when cash is insufficient. Later liability payment does not duplicate expenses. Broader enterprise accounting and native liabilities remain unqualified.
R08	NOT RUN	**Development double-counting trap.** Apply development capitalization and amortization consistently without duplicate cash costs or accounting expenses. *Scope:* No corresponding regression result attached; module existence is not a pass.
R09	NOT RUN	**False profit-funded investment.** Identify opening-equity or borrowing funding honestly instead of calling it earned-profit-funded investment. *Scope:* No corresponding regression result attached; module existence is not a pass.

ID	Coverage	Requirement and exact qualification limit (continued)
R10	PARTIAL	**Failed delivery and refund.** Failed acceptance or unavailable paid delivery must retain costs and trigger the declared refund or entitlement remedy without unearned release. *Scope:* Semantic failure prevents settlement and delivery, retains costs and refunds escrow. A mutated scientific certificate is rejected; conforming delivery survives reconnect. Remote delivery unavailability, external remedies and complete AI-resource costs were not exercised.
R11	NOT RUN	**No successor advantage.** Withhold successor-retention advantage without incremental accepted economic value; preserve valid base-business findings. *Scope:* No corresponding regression result attached; module existence is not a pass.
R12	NOT RUN	**Blue sky hold, positive commercial probe.** Preserve positive commercial side-probe evidence and Blue Sky HOLD together; enforce the side cap and protected instrument/discovery/publication budgets. *Scope:* No corresponding regression result attached; module existence is not a pass.
R13	NOT RUN	**Resource limit with protected completion capacity.** Reject discretionary work exceeding finite caps or completion reserves; close out while preserving failures, liabilities and incomplete claims. *Scope:* No corresponding regression result attached; module existence is not a pass.
R14	NOT RUN	**Resource quantities $\times$ rates, without fictional overhead.** Reconcile justified quantities times rates; remove fictional overhead, retain genuine uncertain costs and prevent double charging. *Scope:* No corresponding regression result attached; module existence is not a pass.
R15	PARTIAL	**Free tool versus equivalent completed service.** Compare equivalent completed services, including supported residual effort for free tools and genuinely negligible-effort free alternatives. *Scope:* Equivalent completed outcomes are compared at hypothetical outside efforts of 0, 300 and 1000 cents. A 500-cent service attracts 0, 0 and 32 of 64 hypothetical buyers. These are assumed analytical choices, not calibrated buyer behavior or measured integration effort.
R16	PARTIAL	**Unsupported frictionless diy and invented buyer friction.** Reject unsupported frictionless DIY as a primary assumption and invented buyer friction used to suppress a free competitor. *Scope:* Zero-effort and positive-effort free alternatives are explicitly hypothetical stress cases. Neither qualifies profitability. Empirical activity-level friction and an adversarial invented-friction rejection policy are unqualified.
R17	PARTIAL	**Preserved originals and untouched confirmation.** Preserve original runs and corrections; require frozen assumptions and untouched confirmation rather than repricing inspected cases. *Scope:* Original development failures, corrected expectation, frozen confirmation, original exchange source and failed refund regression are retained. The specific adversarial commercial-repricing and relabeled-confirmation regression was not executed.
R18	NOT RUN	**Earned-cash-funded capability with fresh delivery benefit.** Trace earned collections through costs/protections into actual new capability construction, then test fresh accepted delivery benefit. *Scope:* No corresponding regression result attached; module existence is not a pass.

ID	Coverage	Requirement and exact qualification limit (continued)
R19	PARTIAL	**Fresh campaign, no inherited shortlist or settings.** Diagnose history without inheriting application choices/settings; compare three fresh downstream possibilities while preserving research continuity. *Scope:* Predecessor diagnosis and three fresh possibilities are separately recorded: reaction intervention, fault isolation and environmental inverse attribution. All downstream candidates remain HOLD. No adversarial inherited-default regression or physical application validation was run.
R20	NOT RUN	**Small profit without a consequential enabling path.** Treat small profitable routine services as commercial evidence only; judge a small enabling transaction by its consequential causal path. *Scope:* No corresponding regression result attached; module existence is not a pass.
R21	NOT RUN	**Earlier candidate reopening and post-selection reuse.** Require material new evidence for reopening; select reuse after the instrument objective or product admission and prohibit false new-invention claims. *Scope:* No corresponding regression result attached; module existence is not a pass.
R22	PARTIAL	**Buyer declines prefunding.** Allow refusal of prefunding for liquidity/trust reasons, retain lost conversion and costs, and never compel participation or inject buyer funds. *Scope:* An unfunded buyer cannot fund; a seven-cent preview expense and liability remain. Hypothetical sensitivity permits nonpurchase. Trust, preference, acquisition expense and actual conversion are not calibrated.
R23	PARTIAL	**Nonpayment, failed funding, and false receipts.** Withhold full useful output for unfunded, pending, forged or failed payment unless frozen assurance or qualified bounded credit passes. *Scope:* Actual bytes remain withheld before qualified local settlement; forged receipts and interrupted funding create no entitlement. No authenticated external payment receipt, remote pending-payment protocol or production custody path was tested.
R24	PARTIAL	**Preview leakage and release-bypass attempts.** Test preview, URL, API, cache, log and key bypasses; preserve value-leakage failures and their economics instead of accepting a cosmetic paywall. *Scope:* Local preview and artifact access enforce the tested entitlement boundary. HTTP URLs, caches, logs, decryption keys and side-channel leakage were not implemented or exhaustively attacked.
R25	PARTIAL	**Independent acceptance and two-sided dispute.** Bind independent acceptance to order/artifact; test buyer strategic rejection, seller misstatement, substitution, validator error and inconclusive dispute. *Scope:* A separate callback binds order, artifact and semantic acceptance, rejects substitution, prevents unsupported buyer veto and withholds earning on inconclusive checks. Roles share one trusted controller; external independence, validator-error calibration and complete dispute service are absent.
R26	PARTIAL	**Duplicate settlement, refund, and concurrent events.** Replay/race settlement and refund events; enforce one economic effect per event, no double charge/spend and valid legitimate reversing entries. *Scope:* Local transactions check identical replay, conflicting keys, refund replay and a settlement/refund race through separate connections. The scientific replay has no second settlement. Distributed retries, provider reconciliation and cross-repository billing remain unqualified.

ID	Coverage	Requirement and exact qualification limit (continued)
R27	PARTIAL	**Interruption at payment and delivery boundaries.** Interrupt/restart at funding, acceptance, settlement, entitlement and refund boundaries; retain obligations and prevent premature or duplicate effects. *Scope:* Injected precommit interruptions roll back funding or settlement/entitlement. Reconnect preserves committed scientific retrieval. This is local transactional recovery, not exhaustive lifecycle interruption or distributed atomicity.
R28	PASS LOCAL	**Escrow is not reinvestable profit.** Keep escrow1000 restricted; enterprise cash100 minus obligations70 and reserve30 leaves investment200 unfunded until valid earned release. *Scope:* Restricted-cash checkpoint: escrow 1000, seller cash 100, deferred obligations 70, reserve 30, earned revenue 0, free cash 0, investment shortfall 200. The helper privately produced and verified before the check; work is unsettled with no earned transfer or buyer entitlement. This narrow pass does not cover the entire unperformed-work then post-release sequence.
R29	PARTIAL	**Chargeback or delay after acceptance.** Account for settlement delays and post-acceptance chargebacks without automatically erasing earned revenue; reassess fees, losses, reserves and funding. *Scope:* Delay and chargeback fixtures retain earned revenue, reversals, fees, losses and unpaid obligations. Late payment reconciles collections without duplicate expense. Original refund failure remains. External chargeback risk, reserve policy and complete commercial costs are unqualified.
R30	PARTIAL	**Justified credit exception and default.** Execute qualified bounded post-delivery credit and late/default accounting; reject excess exposure and never equate receivables with collected cash. *Scope:* Configured credit permits local bytes, separates receivables and cash, caps exposure, rejects ineligible buyers and records default; another fixture collects partially. Eligibility and cap are authored inputs; empirical underwriting, financing and external recovery are unqualified.
R31	NOT RUN	**Protected payment, uneconomic participation.** Preserve successful nonpayment control while withholding Commercial GO if fees, distrust, delay or lock-up defeat participation/economics. *Scope:* No corresponding regression result attached; module existence is not a pass.
R32	NOT RUN	**Frozen mechanism, fresh paired evaluation, funded construction.** Freeze mechanisms and compare fresh paired scenarios with separate ledgers; qualify funding only through actual construction and fresh accepted use. *Scope:* No corresponding regression result attached; module existence is not a pass.
R33	PARTIAL	**Repository preflight without application capture.** Inspect all four governance/ownership sources before changes without adopting an old demo; preserve separate instrument and downstream decisions. *Scope:* Four source/governance trees and active ownership were inspected at exact commits. Historical source convenience did not select the application. This is not an installed compatible release or an executable application-capture adversarial regression.
R34	NOT RUN	**Detached demo does not qualify as integration.** Withhold installed-native integration from detached scripts/demos, regardless of usefulness, profit, repository placement or ecosystem terminology. *Scope:* No corresponding regression result attached; module existence is not a pass.

ID	Coverage	Requirement and exact qualification limit (continued)
R35	PARTIAL	**Exact-source installation and stale evidence.** Reject stale source/install/dependency qualification; bind passes to actual matching loaded bytes and recheck affected changed versions. *Scope:* Source/governance inspection or historical-fixture component checks only; requirement regression and installed integration remain unrun.
R36	NOT RUN	**Relevant non-reuse and irrelevant modules.** Permit justified component non-reuse without omitting mandatory repository roles/security; retain adapter tests and honest native status. *Scope:* No corresponding regression result attached; module existence is not a pass.
R37	PARTIAL	**Additive ownership and release boundaries.** Preserve active ownership and required review/test/signature/publication boundaries; deliver scoped work without destructive or unauthorized release. *Scope:* Ownership/release sources inspected and respected; adversarial regression fixture unrun.
R38	NOT RUN	**Discovery instrument can be built before downstream go.** Allow bounded authorized instrument construction before downstream GO or profit; keep downstream resources and improvement claims separate. *Scope:* No corresponding regression result attached; module existence is not a pass.
R39	NOT RUN	**Instrument operation without discovery improvement.** Preserve functional operation but withhold discovery improvement when unchanged acceptance establishes no additional investigation or productive action. *Scope:* No corresponding regression result attached; module existence is not a pass.
R40	NOT RUN	**Matched-resource predecessor, library, prompt, and ablations.** Reject superiority from unequal resources or omitted strong controls; isolate instrument value with predecessor/library/prompt/ablation comparisons. *Scope:* No corresponding regression result attached; module existence is not a pass.
R41	NOT RUN	**Historical experience is not fresh confirmation.** Retain historical specialist training but reject leaked/inspected confirmation; require independent untouched tasks and version-bound evaluation. *Scope:* No corresponding regression result attached; module existence is not a pass.
R42	NOT RUN	**Source-based hold diagnosis without automatic reversal.** Preserve valid predecessor HOLDs and diagnose missing investigative means from sources; changed status labels alone cannot establish improvement. *Scope:* No corresponding regression result attached; module existence is not a pass.
R43	PARTIAL	**Four-repository mission and returned-evidence trace.** Trace one mission through actual Alpha/Fleet and appropriate separate Jobs/Manager routes with source, artifact, acceptance, cost and return bindings. *Scope:* Source/governance inspection or historical-fixture component checks only; requirement regression and installed integration remain unrun.
R44	PARTIAL	**Imported history does not grant authority.** Treat imported missions/receipts as evidence only; require current authority for execution, checking, settlement and publication at each crossing. *Scope:* Source/governance inspection or historical-fixture component checks only; requirement regression and installed integration remain unrun.
R45	PARTIAL	**Agialpha and usdc are not interchangeable or serial payments.** Reject ambiguous cross-route units, balance netting and serial duplicate settlement; preserve distinct AGIALPHA/USDC identity, custody and authority. *Scope:* Source/governance inspection or historical-fixture component checks only; requirement regression and installed integration remain unrun.

ID	Coverage	Requirement and exact qualification limit (continued)
R46	PARTIAL	**Cross-repository retry does not duplicate execution or billing.** Replay cross-repository dispatch/completion/check/receipt events; prevent duplicate work/billing while charging genuine distinct attempts once. *Scope:* Retry interfaces inspected; no cross-repository duplicate execution/billing fixture run.
R47	PARTIAL	**Fleet execution and checking require separate qualification.** Require separately qualified execution and checking authority, disclose correlated checks, and reject self-authorized or uninformed acceptance. *Scope:* Separate-role source requirements inspected; no qualified Fleet worker/checker pair exercised.
R48	PARTIAL	**Unsupported native edge and bounded adapter.** Record unsupported native edges and bounded adapter assumptions/tests; preserve partial results without simulating away authority or claiming integration. *Scope:* Actual inspector rejected the new envelope shape; source-contract check only, not installed integration or validity of the new instrument.
R49	PARTIAL	**Dev and writer use the same operating implementation.** Investigate the Dev actual source-bound implementation, independently check it and identify corrections/alternate arms with their own fresh evidence. *Scope:* Actual Dev PR5012 source was inspected at its exact head. This fixed-suite compiler is separately frozen and freshly checked; manuscript numbers were audited. The attribution-substitution adversarial test and installed Dev/Writer identity remain unrun.
R50	NOT RUN	**Activation of an old asset is not a newly funded invention.** Distinguish old-asset activation from new funded invention; require reconciled funding, actual construction, new executable artifact and fresh useful use. *Scope:* No corresponding regression result attached; module existence is not a pass.
R51	PARTIAL	**Instrument research, downstream hold, and publication remain distinct.** Allow an honest instrument edition despite downstream HOLD, block held-service spending disguised as research, and preserve independent stage decisions. *Scope:* The completed instrument edition preserves separate HOLD and unqualified states for downstream admission, full-cost profitability, profit-funded successor and native integration. Finalization resource overruns are disclosed. The adversarial attempt to divert instrument resources to a disguised held service was not executed.
R52	PARTIAL	**Publication output and authorized reading links.** Require same-edition LaTeX/PDF/browser output, inspected content and verified references; withhold public publication without authorized served-link checks. *Scope:* Same-edition LaTeX, PDF, browser HTML and source/evidence package have local build, visual-inspection and content-consistency records. Saved artifact-reading links are provided separately. No public publication, external release URL, repository release or publication authority is implied. The full mismatch/unverified-release adversarial regression remains unrun.

D What would qualify the next construction

A successor campaign should begin from an actual unresolved cross-action pair in a consequential setting. It should identify an experiment that is physically or institutionally feasible, establish the new primitive's semantics and error envelope, and retain matched conventional goal-oriented design as a baseline. A field test should include withheld mechanisms or operating regimes, failures, and acquisition costs. An observed benefit must then be translated into an accepted deliverable, fair

completed-service alternatives, and a protected exchange that both parties voluntarily accept.

Native implementation requires a supported extension at the observed schema boundary, compatible installed sources, current execution and checking authority, and separately qualified AGIALPHA or USDC work routes. Financial qualification requires the full activity ledger and actual simulated performance under frozen market assumptions. Any next capability claimed as profit-funded must be constructed after its funding becomes available, with the construction resources, new executable artifact, and fresh productive invocation traced together. These are explicit remaining empirical conditions, not promises of a favorable result.